DAY 1 · FRIDAY 16 OCTOBER 2026 · MFB 545

Fintech, or the unbundled bank

A universal bank sells six services in a single bundle: account, payment, credit, savings, investment and compliance. Fintechs, bigtechs and decentralised protocols take them over one by one. This day takes the bundle apart: your own account openings examined step by step, the unbundled value chain, a radar of a dozen Swiss, European and African players, the numbers of Wave and TWINT redone by hand — then the launch of your project.

🕘 **09:00–12:15 · 13:30–16:45 · four blocks of 90 min** | 💻 **laptop · 📱 phone**

🎤 **the talk is short — 🛠 most of the time, you are the one working**

BY THE END OF DAY 1, YOU WILL BE ABLE TO

- Describe the six building blocks of the banking value chain, what each one earns, and why new entrants attack first the ones that require the least capital.
- Explain, step by step, how a neobank identifies a customer without a branch (reading the ID document, liveness detection, face matching, screening, declarations).
- Document a fintech player on a standard profile: problem solved, customer, revenue, funding, regulator.
- Calculate an average revenue per user (ARPU), a contribution margin and a break-even point, and compare two payment models.
- Formulate, in a team of three, the real customer problem your fintech will solve.

BEFORE YOU START — TICK

- ☐ **Signed in** on this site (the button at the bottom right): what you do today follows you and reaches the instructor.
- ☐ **A spreadsheet open** (Excel or Google Sheets) for Block 3, and a browser for the research in Block 2.
- ☐ **A Google account** created (Google Colab is used on Day 3). Nothing to install.

BLOCK 1 · 09:00–10:30 the unbundled bank · your account openings

1.1 **The unbundled bank**

Six building blocks · why the bundle is coming apart · eKYC in seven steps

🎤 TALK · 25 MIN

1.2 **Opening an account · unbundling a bank**

Your two account openings, step by step · thirteen players, six blocks

🛠 WORKSHOP · 50 MIN

BLOCK 2 · 10:45–12:15 the fintech radar · the quiz

1.3 **Six segments, one standard profile**

The radar · where to find each piece of information

🎤 TALK · 15 MIN

1.4 **“Fintech radar” workshop**

One segment per pair · three players: Swiss, European, African

🛠 WORKSHOP · 45 MIN

1.5 **End-of-morning quiz**

Eight questions · launched in class · counts toward participation

✅ QUIZ · 10 MIN

BLOCK 3 · 13:30–15:00 business models · Wave versus TWINT

1.6 **Business models**

The canvas · five revenue mechanisms · network effects · unit economics

🎤 TALK · 20 MIN

1.7 **Wave versus TWINT**

ARPU, margin, break-even · the simulator

🛠 WORKSHOP · 55 MIN

BLOCK 4 · 15:15–16:45 the “Build your fintech” project · Jeopardy

1.8 **The “Build your fintech” project**

The deliverable, the common thread of the six days, a good problem

🎤 TALK · 15 MIN

1.9 **Your team, your problem**

Teams of three · 2-min pitch · cross-critique · Jeopardy

🛠 WORKSHOP · 35 MIN

← Course page

reset my Day 1 progress on this device

SECTION 1.1 · BLOCK 1

🎤 TALK · 25 MIN

The unbundled bank

Talk by the instructor. If it does not take place, read this section on your own, then do the classification at the bottom of the page.

WHAT A BANK DOES

Six building blocks, sold in a single bundle — and what each one earns

BLOCK 1

Account

Holding the money, keeping the ledger. Earns little in fees; gives access to cheap deposits.

BLOCK 2

Payment

Moving money around: card, transfer, foreign exchange. **Very profitable** (fees, FX margin), almost without capital.

BLOCK 3

Credit

Lending out the deposits: *maturity transformation*. Earns the interest margin; requires capital and a licence.

BLOCK 4

Savings

Savings accounts, retirement provision (the 3rd pillar in Switzerland). Earns the difference between what the bank pays and what it invests.

BLOCK 5

Investment

Advice, wealth management, brokerage. Earns management fees (as a % of assets) and brokerage fees.

BLOCK 6

Compliance

Identifying the customer (KYC), monitoring flows (AMLA), reporting. A cost — which regtechs now sell as a service.

The universal bank **cross-subsidises one block with another**: the “free” account is paid for by payment fees, the FX margin and the interest margin. A new entrant does not need to do everything: it takes *the block that is most profitable for the capital it requires* — historically payments and foreign exchange — and does it better, cheaper, on a phone.

WHY THE BUNDLE IS COMING APART NOW

Three mechanisms

- **The cost of distribution collapses.** A branch costs rent, tellers and opening hours; an app, a server. Remote account opening (below) removed the last mandatory visit to a

branch.

- **The blocks can be bought.** A start-up rents a banking licence and an account from a partner bank (**banking-as-a-service**), identification from a regtech, card processing from a processor, hosting from a cloud — each block through an *API* (Day 2). It only writes the customer journey.
- **Regulation becomes proportionate.** In Switzerland, the *fintech licence* (art. 1b of the Banking Act, 2019) allows a firm to accept deposits from the public up to 100 million francs, provided it neither lends them out nor pays interest — with lighter requirements and no deposit insurance. In the EU, the payment institution (PSD2) and electronic money statuses; in the WAEMU, the BCEAO's **electronic money issuer** licence.

WHO TAKES OVER THE BLOCKS

Three kinds of new entrants — and the bank that remains

- **Specialised fintechs**: one block, one journey, one app (Wise for foreign exchange, Wave for mobile transfers, Klarna for buy-now-pay-later).
- **Bigtechs**: Apple, Google, Alibaba, Tencent. Their advantage is the loop the Bank for International Settlements calls **DNA** — *Data* (the platform's data), *Network* (hundreds of millions of users already there), *Activities* (the financial service is grafted onto commerce, messaging, the phone). Each turn of the loop enriches the data, which improves the service, which attracts users (Frost et al., 2019).
- **Decentralised protocols**: lending, exchange, stablecoins run by programs on a blockchain, with no company in the middle (Day 4).
- **And the bank**: it keeps what requires a balance sheet — credit, large deposits — and often becomes the invisible supplier of fintechs. The opposite movement also exists: *super-apps* (Alipay, M-Pesa) re-bundle everything into a single app (Day 6).

THE MECHANISM · OPENING AN ACCOUNT WITHOUT A BRANCH

Remote identification (eKYC), step by step

In 1.2, you will compare your own account openings with these seven steps. In Switzerland, the framework is **FINMA Circular 2016/7** "Video and online identification". Each step answers a question: *is the document genuine, is the person alive, is it the same person, are we allowed to accept them?*

THE SEVEN STEPS OF AN ONBOARDING

1 · Capturing the document. The app photographs the passport or ID card. It reads the *machine-readable zone* (the two lines of chevrons "<<<" at the bottom of the photo page) by optical character recognition. Each field there carries a **check digit**: the characters are multiplied by 7, 3, 1, 7, 3, 1… (letters A = 10 … Z = 35), the products are added up, and the remainder of the division by 10 is kept. For the number of the ICAO specimen passport, L898902C3, the check digit is 6. A misread or a clumsy forgery shows up immediately.

```
L=21 8 9 8 9 0 2 C=12 3 · weights 7 3 1 7 3 1 7 3 1 →
147+24+9+56+27+0+14+36+3 = 316 → 316 mod 10 = 6
```

2 · Authenticity of the document. The security features are checked in the image: holograms that change with the tilt angle (the user is asked to move the document), microprinting, consistency of the font. Biometric passports also have a **chip** readable by NFC on some phones: it contains the data and the photo, signed by the issuing State — a signature the app can verify.

3 · Selfie and liveness detection. A photo is not enough: someone could present a photo of someone else. *Liveness* detection is *active* (turning the head, following a dot, reading digits aloud) or *passive* (a model analyses skin texture, reflections and depth in a single image).

4 · Face matching. A neural network turns each face — the one on the document and the one in the selfie — into a vector of a few hundred numbers. The closeness of the two vectors is measured; above a **threshold**, it is the same person. High threshold: fewer fraudsters accepted, more genuine customers rejected. Low threshold: the opposite. The threshold is a business choice, not a technical truth.

5 · Screening. The name and date of birth are compared with sanctions lists (SECO, UN, EU, the US OFAC) and with lists of **politically exposed persons**. Names are spelled in several ways: the matching is fuzzy, and a human decides the doubtful cases (Day 5).

6 · Declarations. The customer declares the beneficial owner of the funds (in Switzerland, "Form A" of the banks' Agreement on due diligence), their tax residence (automatic exchange of information) and whether they are a *US person* (FATCA). A questionnaire on the source of funds is added above certain amounts.

7 · The first transfer. For *online* identification (without a video call), FINMA Circular 2016/7 requires a transfer from a bank account in the customer's name: a second bank

has already identified the person, and the name must match. In *video* identification, a trained employee conducts the live call instead.

✓ All 7 steps are shown

> **The ten-minute account is no less checked than the branch one: *the checking has been automated, and each step has an error rate that can be tuned*.**

CHECK · 3 MIN

Specialised fintech, bigtech, decentralised protocol or bank?

▸ CLASSIFY EACH PLAYER

Wise — international transfers at the mid-market rate plus displayed fees

Fintech | Bigtech | Decentralised | Bank

Apple Pay — paying with your iPhone

Fintech | Bigtech | Decentralised | Bank

Uniswap — exchanging tokens on a blockchain

Fintech | Bigtech | Decentralised | Bank

PostFinance — accounts, payments, savings, investment

Fintech | Bigtech | Decentralised | Bank

Alipay — payment built into Alibaba's platform

Fintech | Bigtech | Decentralised | Bank

Wave — mobile wallet in Senegal and Côte d'Ivoire

Fintech | Bigtech | Decentralised | Bank

Aave — lending and borrowing crypto-assets

Fintech | Bigtech | Decentralised | Bank

0 of 7 sorted

SECTION 1.2 · BLOCK 1

🛠 WORKSHOP · 50 MIN

Your account openings · unbundling a bank

A · 20 MIN · ALONE, THEN IN PAIRS

Your own account openings, examined against the seven steps

Everyone has already opened at least one account: in Switzerland on arrival (a bank, PostFinance, an app), and in their home country (a bank account, Wave, Orange Money, MTN MoMo…). **Choose two of these journeys** and reconstruct them from memory, step by step, using the seven eKYC steps seen in 1.1. Nothing to open, no personal data to enter: only what you remember. **10 minutes alone**, then **10 minutes in pairs**: compare your tables and find the step that took the longest.

STEP	Journey 1 — e.g. bank in	Journey 2 — e.g. Wave, D
Appointment or mandatory branch visit	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
1 · ID document photographed or read	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
2 · Authenticity of the document checked	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
3 · Selfie or video call	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
4 · Face compared with the document	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
5 · Screening (sanctions, exposed persons) — often invisible	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
6 · Declarations: beneficial owner, tax residence, source of funds	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know
7 · First deposit or transfer required	✓ Yes ✗ No ? I don't know	✓ Yes ✗ No ? I don't know

0 of 16 cells filled in

Copy the table into my notebook ↓

1. Your two journeys: the table, then for each one the total duration (from first contact to the first possible payment), the documents requested and the fees.

The “Copy the table” button inserts your answers; complete duration, documents and fees.

▸ THE TYPICAL JOURNEY AT A TRADITIONAL BANK, FOR COMPARISON

STEP	WHERE	TYPICAL TIME
Book an appointment	phone, website	a few days
Interview, original ID document copied, Form A (beneficial owner), tax declarations	branch	45–60 min on site
Internal checks, account opened	head office	1–3 working days
Card and codes sent by post	mail	5–10 working days

Typical journey based on the “open an account” pages of Swiss banks; it varies by institution and customer profile. The legal steps are the same as in the app: only the way they are carried out changes.

2. In your two journeys, where was the friction? Which of the seven steps could an app have removed or sped up — and which not?

e.g. In Switzerland, it took a proof of address and a residence permit, three weeks; at Wave, an ID document at the agent’s and ten minutes…

B · 30 MIN · “BANKING VALUE CHAIN” SIMULATOR

Thirteen real players. For each one, the bank's building block it has taken over.

Look up the ones you do not know (company website, two minutes each). Several do a bit of everything: choose *the block that generates their revenue*. A wrong answer greys out the button; the explanation appears when you are right.

▸ ACCOUNT · PAYMENT · CREDIT · SAVINGS · INVESTMENT · COMPLIANCE

TWINT (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Wise (United Kingdom)

Account | Payment | Credit | Savings | Investment | Compliance

Flutterwave (Nigeria)

Account | Payment | Credit | Savings | Investment | Compliance

neon (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Klarna (Sweden)

Account | Payment | Credit | Savings | Investment | Compliance

swisspeers (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Baobab (Africa, formerly MicroCred)

Account | Payment | Credit | Savings | Investment | Compliance

VIAC (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

frankly (Switzerland, Zurich Cantonal Bank)

Account | Payment | Credit | Savings | Investment | Compliance

True Wealth (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Trade Republic (Germany)

Account | Payment | Credit | Savings | Investment | Compliance

PXL Vision (Switzerland)

Account | Payment | Credit | Savings | Investment | Compliance

Smile ID (Africa)

Account | Payment | Credit | Savings | Investment | Compliance

0 of 13 sorted

DEBRIEF · 15 MIN

Three questions for the room: which block has the most new entrants, and why? Which one has almost none? What does the universal bank still do that nobody has taken from it?

SECTION 1.3 · BLOCK 2

🎤 TALK · 15 MIN

Six segments, and how to document a player

Talk by the instructor, or self-study.

THE RADAR'S SEGMENTS

The market cut into six — with, for each, the mechanism that makes money

SEGMENT 1

Payments

Fee per transaction, FX margin. *TWINT, Wise, Stripe, Wave, Flutterwave.*

SEGMENT 2

Credit

Interest margin or merchant fee (buy-now-pay-later). *Klarna, swisspeers, Baobab.*

SEGMENT 3

Savings and wealth management

Management fees as a % of assets, brokerage fees. *VIAC, True Wealth, Yuh, Trade Republic.*

SEGMENT 4

Insurtech

Premiums, distribution commissions; often a traditional insurer's policy sold differently. *Lemonade, Wefox, Turaco.*

SEGMENT 5

Regtech

Subscription or price per check, sold to institutions. *PXL Vision, IDnow, Smile ID.*

SEGMENT 6

Infrastructure and BaaS

Price per account or per API call, sold to other fintechs. *Solaris, Swan, Onafriq, Paystack.*

The names are starting points, not a closed list: a player you know from your own country is worth more than a name from this page.

THE STANDARD PROFILE

Five questions, and where to find each answer

- **Problem solved** — in one sentence, from the customer's point of view, *without* the name of the technology. Source: the home page, then customer reviews (app stores).
- **Customer** — who pays, and who uses (not always the same: at TWINT the user pays nothing, the merchant pays). Source: the price list.

- **Revenue** — the exact mechanism: “1% on transfers”, “0.5% of assets per year”, “CHF 9 per month”. Source: the pricing terms, the annual report if published.
- **Funding** — funds raised, investors, profitable or not. Source: press releases, specialised press (Finextra, TechCabal, Fintech News Switzerland), Crunchbase.
- **Regulator** — who authorises it, under which status. Source: the regulator’s register (FINMA: list of authorised institutions; BCEAO: list of electronic money issuers; EU: national registers and the EBA register), and the website’s legal notice.

> **A profile is judged on its revenue mechanism. *“Makes money from fees” is not an answer; “charges 1% on every transfer, deposits and withdrawals free” is one.***

CHECK · 3 MIN

Where to look?

› WEBSITE AND PRICING · REGULATOR’S REGISTER · PRESS AND RELEASES · ANNUAL REPORT

How much it costs to send 100,000 FCFA with a mobile money provider

Website and pricing | Regulator’s register | Press, releases | Annual report

Whether a Swiss fintech has a banking licence, a fintech licence, or nothing

Website and pricing | Regulator’s register | Press, releases | Annual report

How much a fintech raised in its last funding round, and from whom

Website and pricing | Regulator’s register | Press, releases | Annual report

The revenue and profit of a listed fintech

Website and pricing | Regulator’s register | Press, releases | Annual report

Whether an institution is licensed to issue electronic money in Côte d’Ivoire

Website and pricing | Regulator’s register | Press, releases | Annual report

0 of 5 sorted

SECTION 1.4 · BLOCK 2

🛠 WORKSHOP · 45 MIN

“Fintech radar” workshop

30 MIN · IN PAIRS

One segment per pair. Three real players: one Swiss, one European, one African.

Nine people: three pairs and one group of three. Each group takes a different segment from the six in 1.3 (the two remaining segments are for groups that finish early). Fill in one profile per player: the button inserts the template into an empty field. Everyone fills in their own three profiles — it is your notebook, and it is saved.

Insert the template into the empty fields

Swiss player — segment, name, and the five lines of the profile

European player

African player

15 MIN · REPORT-BACK AND COLLECTIVE RADAR

Three minutes per group. The radar is built on the screen.

Each group presents its segment in three minutes: the three players, and *one* thing they have in common (the same revenue mechanism? the same customer?). The whole room's profiles are shown on the screen, anonymously, during the report-back.

Report-back 2:55 Pause ↺

SECTION 1.5 · BLOCK 2

✅ QUIZ · 10 MIN

End-of-morning quiz

INTERACTIVE QUIZ · COUNTS TOWARD PARTICIPATION

Eight questions, one attempt each, all explained

Launched in class, so that everyone starts at the same time; this page unlocks by itself. Interactive quiz results count toward the 10% participation mark; a missed quiz scores zero unless arranged in advance. On your own, without an AI assistant.

0 / 8 answered

1. “Unbundling” the universal bank means…

- merging several banks into one
- closing branches in favour of an app
- having specialised players offer separately services that the bank used to sell in a single bundle
- abolishing banking regulation

2. The Swiss fintech licence (art. 1b of the Banking Act) allows a firm…

- to accept deposits from the public up to 100 million francs, without lending them out or paying interest
- to lend out the deposits received without limit
- to issue banknotes
- to operate without any supervision

3. In Switzerland, opening an account by video or online is governed by…

- the GDPR

- the PSD2 directive
- the MiCA regulation
- FINMA Circular 2016/7

4. neon offers accounts and cards in Switzerland. How, without a banking licence?

- It does not need one to hold accounts
- The accounts are held by a partner bank
- It uses the fintech licence to lend
- The funds are placed on a blockchain

5. A two-sided network effect is…

- a service that costs half as much at large scale
- a network of two banks
- an app available on two operating systems
- two groups that attract each other: the more customers pay with an app, the more merchants accept it, and vice versa

6. ARPU of 500 FCFA per month, variable costs of 250 FCFA per user, fixed costs of 1.5 billion FCFA per month. The break-even point is at…

- 6 million active users
- 3 million
- 12 million
- 1.5 million

7. Wave's pricing model in Senegal is…

- a fixed monthly subscription
- tiered, decreasing fees on each operation
- free deposits and withdrawals, 1% on transfers
- free transfers funded by advertising

8. Which of these players belongs to regtech?

- A low-fee online broker
- A provider of identity verification and sanctions-list screening sold to banks
- A payment app between friends
- A crowdfunding platform

SECTION 1.6 · BLOCK 3

🎤 TALK · 20 MIN

Business models: who pays, how much, and when it becomes profitable

Talk by the instructor, or self-study.

THE CANVAS

Nine boxes to describe how a company makes money

The *Business Model Canvas* (Osterwalder and Pigneur, 2010), filled in here for Wave in broad strokes. On the right, value and revenue; on the left, what it takes to produce them and what it costs.

KEY PARTNERS
Banks where customers' funds are deposited · agent network · telecom operators

KEY ACTIVITIES
App, transaction processing, recruiting and managing agents, compliance

KEY RESOURCES
Electronic money issuer licence, software, liquidity at the agents

VALUE PROPOSITION
Send money from a phone for 1%, deposit and withdraw for free

CUSTOMER RELATIONSHIPS
Simple app, QR card for those without a smartphone, customer service

CHANNELS
Neighbourhood agents, word of mouth, app

CUSTOMER SEGMENTS
Unbanked or underserved individuals · merchants · companies paying salaries

COST STRUCTURE
Agent commissions · technology · salaries · marketing · compliance

REVENUE STREAMS
1% on transfers · services to merchants and businesses · income on deposited funds, within regulatory limits

HOW A FINTECH CHARGES

Five revenue mechanisms

- **Commission or fee per transaction** — a percentage or a fixed amount on each operation (merchant fee, transfer fee).
- **Interchange** — the share of each card payment received by the bank that issued the card. This is how neobanks make money on a "free" card (Day 2).
- **Margin** — on foreign exchange (converting at a worse rate than the mid-market rate) or on interest (lending at a higher rate than one's own funding cost).

- **Subscription** — a monthly price for a premium offer.
- **Management fees** — an annual percentage of assets under management.

NETWORK EFFECTS

Why the first to reach critical mass often wins everything

Direct effect — each new user makes the service more useful to the others. Among n users, the number of pairs that can send each other money is $n(n-1)/2$: 45 pairs for 10 users, 4,950 for 100, 499,500 for 1,000. The value grows much faster than the number of users.

Indirect, or two-sided, effect — two groups that attract each other: the more customers pay with TWINT, the more merchants accept it, the more customers adopt it. The starting problem is the *chicken-and-egg* problem; the usual solution is to **subsidise one side** — free P2P at TWINT, free withdrawals at Wave — and charge the other.

UNIT ECONOMICS

Four formulas for Block 3

ARPU = revenue for the month ÷ active users for the month

Contribution margin per user = ARPU – variable costs per user

Break-even (in users) = fixed costs for the month ÷ contribution margin per user

Take rate = revenue ÷ volume that flows through

Variable costs follow each user (agent commissions, processing, SMS, fraud); *fixed* costs are there even with no users (salaries, technology, compliance, head office). A fintech loses money as long as it is below its break-even point, and each user above it brings in their contribution margin.

CHECK · 3 MIN

Which revenue mechanism?

PER TRANSACTION · INTERCHANGE · MARGIN · SUBSCRIPTION · MANAGEMENT FEES

The merchant pays 1.3% of each purchase paid with the app

Per transaction | Interchange | Margin | Subscription | Management fees

A neobank gives the card away; it receives a small share of each payment made with it

Per transaction | Interchange | Margin | Subscription | Management fees

A bank converts your francs into euros at 0.91 while the mid-market rate is 0.93

Per transaction | Interchange | Margin | Subscription | Management fees

A “premium” offer at CHF 9 per month with travel insurance and free withdrawals

Per transaction | Interchange | Margin | Subscription | Management fees

A robo-advisor charges 0.5% of assets per year

Per transaction | Interchange | Margin | Subscription | Management fees

An online lender lends at 12% and funds itself at 5%

Per transaction | Interchange | Margin | Subscription | Management fees

Wise displays a fee of CHF 4.20 on a transfer, at the mid-market rate

Per transaction | Interchange | Margin | Subscription | Management fees

0 of 7 sorted

SECTION 1.7 · BLOCK 3

🛠 WORKSHOP · 55 MIN

Wave versus TWINT — redoing the numbers

THE DATA · PER ACTIVE USER PER MONTH

Two payment apps, two continents, two models

	WAVE (SENEGAL) · FCFA	TWINT (SWITZERLAND) · CHF
Person-to-person transfers (P2P)	50,000 · fee **1%** for the sender	80 · **free**
Payments at merchants	— (not included)	120 · merchant fee **1.3%**
Cash deposits and withdrawals at agents	60,000 · free for the customer; Wave pays the agent **0.3%**	— (no cash)
Other variable costs (processing, SMS, support, fraud)	70	0.46
Company fixed costs, per month	1.5 billion	5.5 million
Active users	8 million	6 million

⚠ **Stylised figures for the exercise**, built on public orders of magnitude (Wave's "1% on sending, deposits and withdrawals free" pricing; a merchant fee of around 1.3% for TWINT; free P2P) and rounded so the calculations can be done by hand. These companies' real figures are not published in detail: do not quote these as such.

25 MIN · BY HAND, THEN IN THE SPREADSHEET

Five calculations

Do them in your spreadsheet: one column per company, one row per quantity. Enter each result here; a mistake can be corrected as many times as needed.

Wave · ARPU. Revenue per active user per month (FCFA).

… FCFA Check

Wave · contribution margin per user per month (FCFA).

… FCFA Check

Wave · break-even in active users.

… users Check

TWINT · break-even in active users.

… users Check

Price war. A competitor forces Wave to go from 1% to 0.8% on transfers. New break-even point, in active users?

… users Check

10 MIN · VARY THE INPUTS

The simulator: move the sliders, watch the break-even point move

Wave | TWINT

Fee on transfers 1.0 %

Transfers per user per month 50,000

Agent commission (on 60,000 of deposits/withdrawals) 0.30 %

Fixed costs per month (billions) 1.5

Active users (millions) 8.0

500	250	6.00 M	0.50 bn
ARPU · FCFA	MARGIN / USER	BREAK-EVEN · USERS	RESULT FOR THE MONTH · FCFA

Above break-even: each additional user adds their margin to the result.

10 MIN

Wave, TWINT or both?

▸ CLASSIFY EACH FEATURE

Sending money between individuals is free for the user

Wave | TWINT | Both

Depositing and withdrawing cash is free

Wave | TWINT | Both

Relies on a network of physical agents

Wave | TWINT | Both

Debits a bank account directly

Wave | TWINT | Both

Network effect between users and merchants

Wave | TWINT | Both

Payment by QR code

Wave | TWINT | Both

Forced a telecom operator to cut its prices sharply

Wave | TWINT | Both

0 of 7 sorted

✍ **In these data, Wave's ARPU is about CHF 0.70, against CHF 1.56 for TWINT. In three sentences: how can Wave still be profitable, and what would make it fragile?**

e.g. Its fixed costs in CFA francs are low for…; it exceeds its break-even point by…; but cutting its fee to 0.8% would move the break-even point to…

SECTION 1.8 · BLOCK 4

🎤 TALK · 15 MIN

The “Build your fintech” project

Talk by the instructor, or self-study.

40% OF THE GRADE · REPLACES THE MIDTERM EXAM

Two teams of three, one project: a pension-savings app for Senegal, Congo and their neighbours.

The app lets people who have no pension, or a very small one, save for their old age by paying **small amounts every month until 65**, invested in **Swiss blue-chip equity funds**, among the most stable on the market. The app and the company must be **transparent and earn trust**: that is the condition for anyone to hand over their money for thirty years. Each team picks its country (Senegal, Congo, DRC, Côte d’Ivoire…) and its customer. Each day adds a piece to the project, at the end of the afternoon, using what has just been learned. *(Brief set on 24 October 2026.)*

Day 1 · 16 Oct.

The problem and the customer — the team, the market, the problem in one sentence

Day 2 · 23 Oct.

The payment flow and the revenue model

Day 3 · 24 Oct.

The pension-savings plan — payments, fees, capital at 65, transparency

Day 4 · 30 Oct.

The technology choice — is a blockchain needed to prove the money is there? a reasoned decision

Day 5 · 7 Nov.

The regulatory file and the risk map

Day 6 · 14 Nov.

The financing plan · **presentation to the jury** (10 min + 10 min of questions)

The deliverable: (a) a report of at most fifteen pages — problem and customer, market, the savings plan and the customer journey with a clickable prototype, revenue model and unit economics, the payment flow to the Swiss fund and back, what creates trust (custody of the funds, transparent fees, information on risks), reasoned technology choice, regulatory compliance in Switzerland and in the target country, financing plan; (b) the prototype of the customer journey; (c) the oral presentation on 14 November. **Report due Friday 20 November 2026, 23:59**, on this site — like everything else in the project: nothing is handed in on Moodle or the Swiss UMEF platform. One grade for the whole team, weighted where appropriate by an individual assessment of each member’s contribution.

A GOOD STARTING POINT

A problem, not a solution

- **A specific customer**: “self-employed seamstresses in Dakar”, not “Africans”.
- **What they do today**: the current workaround (cash, tontine, a transfer through a cousin) — that is your real competitor.
- **What it costs them**: in money, in time, in risk. A figure, even a rough one.
- **Why now**: what has changed (a phone in every pocket, instant payments, a new regulation).

> **The word “blockchain” or “AI” does not appear in a problem statement. *The technology is chosen on Day 4, once the problem is understood.***

CHECK · 3 MIN

Good problem, disguised solution, or too vague?

▸ CLASSIFY EACH STARTING POINT

“Build a neobank for Africa”

Good problem | Disguised solution | Too vague

“An app with AI and blockchain for payments”

Good problem | Disguised solution | Too vague

“African students in Geneva often pay more than 5% to send CHF 200 to their family every month”

Good problem | Disguised solution | Too vague

“Seamstresses in Dakar have no credit history to borrow 300,000 FCFA of fabric before Tabaski”

Good problem | Disguised solution | Too vague

“Create a digital token to replace the CFA franc”

Good problem | Disguised solution | Too vague

“Swiss SMEs wait on average more than 30 days for their invoices to be paid”

Good problem | Disguised solution | Too vague

“Improve financial inclusion”

Good problem | Disguised solution | Too vague

0 of 7 sorted

SECTION 1.9 · BLOCK 4

WORKSHOP · 35 MIN | PITCH, CRITIQUE AND JEOPARDY · 40 MIN

Your team, your problem

10 MIN · FORMING THE TEAM

Two teams of three, one country each

A team should preferably mix people who do not know each other yet and who do not come from the same country. Each member fills in the field below with the same information.

1. Team name · the three members · the country your pension-savings app targets (Senegal, Congo, DRC, Côte d'Ivoire… — and which city first)

e.g. Team Teranga · A., B., C. · Senegal, Dakar

25 MIN · THE PROBLEM

The problem in one sentence, and the customer in three lines

2. The problem: *For [specific customer], who today [does it how], the problem is [which one], which costs them [how much].*

e.g. For women traders aged 25 to 45 in Dakar, who save today in cash or in a tontine, the problem is that they will have no pension at 65 and their savings earn nothing, which costs them…

3. The customer: who exactly, how many are there (order of magnitude and source), and how do you know the problem exists?

e.g. About … students (source: …); two team members live this problem; three people interviewed during the break…

20 MIN · TWO-MINUTE PITCH AND CROSS-CRITIQUE

Each team presents; the other asks its three questions

Pitch 1:55 Pause ↺

The listening team asks, in this order: **1.** Does the customer really exist, and have you met them? **2.** What do they do today, and why would they change? **3.** What could kill this project (a competitor, regulation, acquisition cost)?

4. The critique received, and what the team changes in its problem.

e.g. We were asked whether students do not already use Wise; we will compare the costs on Day 2…

20 MIN · CLASS GAME

Jeopardy "Fintech — fundamentals"

The three project teams play against each other on the classroom screen: five categories, twenty-five questions on the day. A team picks a square; the first raised hand answers.

Open the Jeopardy (classroom screen) ↗

BEFORE DAY 2 · FRIDAY 23 OCTOBER

- Refine the problem with the critique received; interview two potential customers.
- Day 2 is about payments: note how your customer pays and receives money today — by what means, at what cost, how quickly.